PL-300 Exam Preparation

Session 1: Data & Modelling Masterclass

Learner Guide · Exam Strategy Workshop

📖 **How to Use This Guide This session is not a Power BI lesson — it is an exam strategy workshop.**

You already know Power BI. This guide teaches you how Microsoft writes questions, where the marks are, and how to choose the right answer under exam conditions.

What This Session Covers

Part	Topic
Part 1	Understanding the Passing System
Part 2	Where the Marks Are
Part 3	How Microsoft Thinks
Part 4	Question Types & Real Examples
Part 5	Power Query — Key Concepts
Part 6	Data Modelling — Key Concepts
Wrap-Up	Cheat Sheets to Keep

Part 1 — Understanding the Passing System

The Pass Mark

PL-300 Pass Mark

700 out of 1000

Scaled score · Range: 100–1000

⚠️ **Important** ❌ **Common mistake: "700/1000 means I need 70% of questions correct."**

This is wrong. Microsoft uses weighted scoring. The 700 is a scaled score, not a raw percentage. Some questions are worth more than others, and the exact formula is not published.

Why the Score Is Not a Simple Percentage

- Some questions carry more weight than others.
- Case study questions may count more than standard questions.
- Beta questions are included in the exam but are unscored — you cannot tell which ones they are.
- Different exam versions may use slightly different scoring models.

Rough Rule of Thumb

Based on what experienced PL-300 trainers observe:

Correct Answers (approx %)	Likely Outcome
Below 60%	Usually Fail
65–70%	Borderline — at risk
70–75%	Usually Pass
80% and above	Comfortable Pass

🎯 **Your Target Your target: aim to score 75% or above consistently on practice questions before booking the exam.**

That buffer above the borderline zone protects you on exam day when nerves, tricky wording, or a difficult scenario question costs you marks.

Exam Structure at a Glance

Item	Value
Exam Duration	100 minutes

Number of Questions	~40–60
Passing Score	700 / 1000
Your Revision Target	75%+ consistently on practice questions

Readiness Guide — 90-Question Revision Pack

Score on 90-Question Pack	What It Means
90/90 (100%)	✓ Exam Ready
80/90 (89%)	✓ Very Strong — book with confidence
70/90 (78%)	✓ Likely Pass — one more revision round recommended
63/90 (70%)	⚠ Borderline — more practice needed on weak topics
Below 60/90	✗ More Revision Needed — identify and close your gaps first

Part 2 — Where the Marks Are

The 4 Official Exam Domains

Microsoft publishes domain weightings, not exact marks per topic. The exam has approximately 40–60 questions with a passing score of 700/1000.

Exam Domain	Published Weight	Approx Marks (out of 1000)
Prepare the Data	25–30%	250–300
Model the Data	25–30%	250–300
Visualize and Analyze Data	25–30%	250–300
Manage and Secure Power BI	15–20%	150–200

📌 **Key Point** Three domains carry equal weight (25–30% each). But within Model the Data, DAX and Time Intelligence are hidden inside that 25–30% — making it the most content-dense domain of the four.

Inside "Model the Data" — The Hidden Depth

This is where most learners underestimate the exam. The 250–300 marks for Model the Data break down approximately as:

Subtopic	Approx Share of Domain
Relationships & Star Schema	~30% (75–90 marks)
DAX Measures	~30% (75–90 marks)
Time Intelligence	~15% (35–45 marks)
Date Tables	~10% (25–30 marks)
Storage Modes & Composite Models	~15% (35–45 marks)

Marks Value by Topic — Full Exam View

Topic	Approx Marks Value
Data Modelling (Relationships, Star Schema, Storage)	~90
DAX (CALCULATE, FILTER, ALL, SUMX, RELATED)	~90
Power Query (Transformations, Profiling, Sources)	~80
Visuals (Charts, Slicers, Drillthrough, Bookmarks)	~70
Time Intelligence (SAMEPERIODLASTYEAR, TOTALYTD)	~40

Power BI Service (Workspaces, Apps, Gateways)	~40
Security / RLS	~30
Performance Optimisation	~20

💡 **Key Insight Modelling + DAX ≈ 180 marks. Power Query ≈ 80 marks. Together they account for more than half the exam.**

A learner who masters Relationships, Star Schema, CALCULATE, FILTER, and SAMEPERIODLASTYEAR is already covering 40–50% of the exam scoring potential.

If You Have 5 Hours to Revise — Spend It Like This

Topic	Recommended Time
Modelling + Relationships	25% (~75 min)
DAX	25% (~75 min)
Power Query	20% (~60 min)
Time Intelligence	10% (~30 min)
Visuals	10% (~30 min)
Service + RLS	10% (~30 min)

The "Pass the Exam" Top 12 Topics

Learners who are strong in these 12 topics typically cover 40–50% of the exam scoring potential:

Modelling & Relationships	DAX & Service
✓ Star Schema (Fact vs Dimension)	✓ CALCULATE
✓ Relationships & Cardinality	✓ FILTER
✓ Active vs Inactive Relationships	✓ SAMEPERIODLASTYEAR
✓ Date Tables (Mark as Date Table)	✓ USERELATIONSHIP
✓ Storage Modes (Import / DirectQuery / Dual)	✓ RLS (Row-Level Security)
✓ Composite Models	✓ Workspaces & Apps

Part 3 — How Microsoft Thinks

The Core Shift: Knowledge → Judgement

The PL-300 is not a knowledge test. It is a judgement test.

Microsoft rarely asks:

❌ "What does CALCULATE do?"

Microsoft almost always asks:

✅ "A company has these requirements. They need X and Y. Which solution should you implement?"

The 4-Part Decoding Framework

Before you look at the answer options, identify these four things in every question:

Element	What to look for
Requirement	What does the business actually need?
Constraint	What limits the solution? (cost, mode, platform, performance)
Keyword	The signal word that points to a category (e.g. "previous year" → Time Intelligence)
Best Solution	The answer that meets ALL requirements with no unnecessary complexity

📝 **Practice Tip Practice saying this out loud for the first few questions:**

"The requirement is… the constraint is… the keyword is… therefore the answer is…"

After a few rounds you will internalise the pattern without the verbal scaffold.

The Most Dangerous Wrong Answers

Microsoft deliberately places answers that are correct in isolation but wrong for the scenario:

Trap Type	Example
Technically correct, wrong context	"Use CALCULATE" — correct function, wrong scenario (no filter context needed)
Partially meets requirements	Meets the performance need but ignores the security requirement
Over-engineered solution	Composite model when simple Import mode would do
Real feature, wrong use	"Use a gateway" — gateways handle refresh, not report sharing

Part 4 — Question Types & Real Examples

The PL-300 Big 8 — Where the Marks Are Won

Question Type	Exam Frequency
Data Modelling (Star Schema, Relationships)	⭐⭐⭐⭐⭐ Very High
Relationships (Cardinality, Active/Inactive)	⭐⭐⭐⭐⭐ Very High
DAX (CALCULATE, FILTER, ALL, SUMX)	⭐⭐⭐⭐ High
Time Intelligence (SAMEPERIODLASTYEAR, TOTALYTD)	⭐⭐⭐⭐ High
Power Query (Profiling, Transformations)	⭐⭐⭐⭐ High
HOTSPOT (Complete the blanks / dropdowns)	⭐⭐⭐⭐ High
Power BI Service / Security (RLS)	⭐⭐⭐ Medium
Visuals (Drillthrough, Tooltips, Sync Slicers)	⭐⭐⭐ Medium

⚠️ **Most Common Revision Mistake** **The biggest revision mistake is spending hours memorising visual types and not enough time on relationships, modelling, storage modes, date tables, CALCULATE, and time intelligence.**

Those are the areas Microsoft tests repeatedly because they reflect real-world Power BI skill.

The 18 Question Types — Quick Reference

Type	What Microsoft Tests	Frequency
1. Definition	Simple recall — "Which function returns previous year?"	⭐ Low (5%)
2. Scenario — Single Answer	Requirements + constraints → best solution	⭐⭐⭐⭐⭐ Very High
3. Scenario — Multiple Choice	Choose TWO correct actions	⭐⭐⭐ Medium
4. HOTSPOT	Complete dropdowns / fill blanks in a solution	⭐⭐⭐⭐ High
5. Drag & Drop	Match requirement to solution	⭐⭐ Medium
6. Build List	Arrange steps in correct order	⭐⭐ Low
7. Yes / No	Does this proposed solution meet the goal?	⭐⭐⭐⭐ High
8. Complete the Solution	Finish a DAX or M formula	⭐⭐⭐ Medium
9. Data Modelling	Fact vs dimension, cardinality, filter direction	⭐⭐⭐⭐⭐ Very High
10. Storage Mode	Import / DirectQuery / Dual — which to use?	⭐⭐⭐⭐ High

11. Relationship	Role-playing dimensions, USERELATIONSHIP	⭐⭐⭐⭐⭐ Very High
12. DAX Function	Running total / previous year / remove filters	⭐⭐⭐⭐ High
13. Time Intelligence	Previous month / year / YTD — choose function	⭐⭐⭐⭐ High
14. Visualisation	Sync slicer / drillthrough / tooltip	⭐⭐⭐ Medium
15. Security	Dynamic RLS for row-level filtering	⭐⭐⭐ Medium
16. Power BI Service	App / workspace / gateway	⭐⭐⭐ Medium
17. Performance Optimisation	Aggregations / star schema / dual storage	⭐⭐ Medium
18. Case Study	Company background → 5–10 linked questions	⭐⭐⭐⭐ High

📌 **Note** Case Study questions often appear at the start or end of the exam. Read the requirements section carefully before answering any of the linked questions — all answers depend on holding the full company context in mind.

5 Real Questions That Cover 70% of PL-300 Patterns

Question	Pattern	What to Learn
Q14 — Text.End / Text.Replace	HOTSPOT + Complete the Solution	Select the correct M function to reformat phone numbers. Focus on matching the required output to the function name.
Q45 — Previous Quarter Sales	Complete the Solution + Time Intelligence	Finish the CALCULATE formula using SAMEPERIODLASTYEAR or DATEADD. Understand what each time function shifts.
Q153 — Folder with JSON Files	Scenario Single Answer + Data Source	Use the Folder connector, not the JSON connector. The keyword "multiple files in a folder" decides the answer.
Q155 — DirectQuery + Import Table	Storage Mode + Performance	Customer table must use Dual storage to serve both Import and DirectQuery sides of the model.
Q376 — Multiple Date Columns	Yes/No + Data Modelling	Multiple date tables for Order/Due/Delivery dates = role-playing dimensions. Answer: Yes, this meets the requirement.

The Keyword → Category Decoder

As soon as you spot one of these phrases in a question, you know which framework applies:

If you see this in the question…	Think this category
Import + DirectQuery	Composite Model / Dual Storage

Two or more date columns (Order Date, Delivery Date)	Role-Playing Dimensions or USERELATIONSHIP
Previous year / previous quarter / YTD	Time Intelligence
Text cleanup / column splitting / profiling	Power Query
Folder / JSON / CSV / SharePoint	Data Source Selection
"Does this meet the goal?"	Yes/No — evaluate the proposed solution carefully
Fact table + dimension table	Relationships and Modelling
Dataset is slow / performance issue	Star Schema, Aggregations, Dual Storage
Only their own region / department / row	Row-Level Security (RLS)
Running total / prior period comparison	CALCULATE + Time Intelligence

Part 5 — Power Query: Key Concepts

Questions in this section use these question numbers from the bank: Q1, Q3, Q10, Q14, Q18, Q23, Q27, Q83, Q89, Q93, Q103, Q112, Q118, Q153, Q288.

Data Profiling

- Column Profile shows distribution, min, max, and errors for the selected column only (based on 1,000 rows by default).
- Column Quality shows Valid / Error / Empty percentages across all columns at once.
- Distinct count = number of different values. Unique count = values that appear exactly once. These are not the same — this is a common exam trap (Q18, Q89).

⚠ **Common Pitfall** Q18 / Q89 trap: Microsoft asks "which profiling view shows X?" — Column Profile (one column, detailed) vs Column Quality (all columns, summary). Know the difference.

Text Transformations

- Text.End — extracts characters from the RIGHT of a string.
- Text.Start — extracts characters from the LEFT of a string.
- Text.Replace — replaces a specific substring within a value.
- Table.ReplaceValue — replaces values across an entire column.
- Q14 is a HOTSPOT question: you select the correct M function from a dropdown to reformat phone numbers. Read what the output needs to look like, then match the function.

Merge & Join Types

Join Type	What It Returns
Left Outer	All rows from the left table, matching rows from the right
Inner Join	Only rows that match in both tables
Full Outer	All rows from both tables, matched where possible
Left Anti	Rows in the left table with NO match in the right (non-matching rows)
Right Anti	Rows in the right table with NO match in the left

📌 **Note** Q112 exam approach: Microsoft describes the result, not the join name. Map the description to the join type. Example: "keep all customers even if they have no orders" = Left Outer join.

Data Sources & Connectors

- Folder connector — combines all files in a folder in one step. Handles new files added later automatically. Use this when the scenario says "multiple files in a folder."
- SharePoint Folder connector — same principle, for cloud-based SharePoint sources.

- Do not select the specific file-type connector (e.g. JSON) when the question describes a folder of files. The Folder connector is always the correct answer in that scenario (Q153).

Part 6 — Data Modelling: Key Concepts

Questions in this section include: Q11, Q26, Q29, Q31, Q33, Q36, Q65, Q86, Q121, Q124, Q126, Q128, Q142, Q156, Q162, Q175, Q205, Q221, Q244, Q294, Q299, Q302, Q318, Q381, Q426.

4 Questions to Ask Before Every Modelling Question

Ask Yourself	Why It Matters
Is this a Fact or Dimension table?	Determines star schema design, grain, and relationship direction
What is the cardinality?	One-to-many vs many-to-many vs one-to-one — determines join type
Do we need multiple date roles?	Signals role-playing dimensions or USERELATIONSHIP
Import, DirectQuery, or Dual?	Determines refresh model, performance, and composite model design

Star Schema

- Fact table = transactions, events, measurements. Contains foreign keys and numeric values. Examples: Sales, Orders, Bookings.
- Dimension table = descriptive data. Contains the primary key and attributes. Examples: Customer, Product, Date, Region.
- Every relationship flows from dimension (one side) → fact (many side). Never dimension → dimension.

💡 **Key Concept Rule of thumb:**

"If every row represents one transaction, it is a fact table. If every row represents one entity (one customer, one product), it is a dimension table."

Relationships — Cardinality & Filter Direction

- One-to-Many (*:1) — the default. Dimension primary key → fact foreign key.
- Many-to-Many (*:*) — requires a bridge table or bidirectional filtering. Use only when genuinely needed.
- Bidirectional cross-filter — filters flow both ways. There is a performance cost. Only use when required.

Active vs Inactive Relationships

- Only ONE active relationship can exist between two tables at a time.
- Inactive relationships are needed when a second relationship exists (e.g. Order Date AND Delivery Date both linking to a Date table).
- USERELATIONSHIP activates an inactive relationship inside a CALCULATE expression.

⚠️ **Common Pitfall** Q36 / Q65 trap: you do not need to delete the active relationship to create a new one. The correct approach is always to create an inactive relationship and use USERELATIONSHIP in the DAX measure.

📋 **Formula Reference USERELATIONSHIP formula pattern:**

Delivery Revenue = CALCULATE ([Total Revenue], USERELATIONSHIP (Sales[DeliveryDate], DateTable[Date]))

This activates the inactive relationship instead of the active one for this calculation only.

Storage Modes — Import, DirectQuery, Dual

Storage Mode	Behaviour & When to Use
Import	Data cached in memory. Fast queries. Needs scheduled refresh. The default choice.
DirectQuery	Queries sent live to the source. Always current. Slower. Limited DAX support.
Dual	Behaves as Import when queried alone; as DirectQuery when queried alongside a DirectQuery table. Used for dimension tables in composite models.
Live Connection	Connects to Analysis Services or a published Power BI dataset. No local model.

💡 **Key Concept Dual storage rule:**

"If the table is a dimension used in a mixed Import/DirectQuery composite model, set it to Dual."

Microsoft uses this scenario repeatedly in the exam.

Date Tables & Role-Playing Dimensions

- A proper Date table must be contiguous (no date gaps), cover the full data range, and be marked as a Date Table in Power BI.
- Marking as a Date Table is required for time intelligence functions (TOTALYTD, SAMEPERIODLASTYEAR) to work correctly.
- Role-playing dimensions: when multiple date fields exist (Order Date, Delivery Date, Due Date), create separate Date tables or use inactive relationships + USERELATIONSHIP.
- Q376 (Yes/No): creating three separate Date tables meets the requirement for role-playing date dimensions. Answer: Yes.

📌 **Decision Tree Decision tree:**

ONE date column in the fact table → one Date table with one active relationship.

TWO OR MORE date columns → role-playing dimensions (separate Date tables or inactive relationships + USERELATIONSHIP).

Wrap-Up — Cheat Sheets to Keep

Cheat Sheet A — Modelling Rules

Topic	The Rule
Fact Table	Each row = one transaction / event. Contains foreign keys and numeric measures.
Dimension Table	Each row = one entity. Contains the primary key and descriptive attributes.
Active Relationship	The default relationship used in all DAX calculations automatically.
Inactive Relationship	A second relationship between the same tables. Activated by USERELATIONSHIP inside CALCULATE.
Composite Model	A model containing both Import and DirectQuery tables.
Dual Storage Mode	Set on dimension tables in a composite model — behaves as Import or DirectQuery depending on context.
Role-Playing Dimension	Multiple date columns (Order/Delivery/Due) each linking to a separate Date table or inactive relationship.
Mark as Date Table	Required for time intelligence functions. Date column must be contiguous with no gaps.
Many-to-Many (*:*)	Requires a bridge table or bidirectional filtering. Use only when necessary.
Bidirectional Filter	Filters flow both ways. Performance cost. Use only when required.

Cheat Sheet B — Question Type Decoder

If the question contains…	The answer category is…
Import + DirectQuery in same model	Composite Model → dimension tables use Dual storage
Two date columns in fact table	Role-Playing Dimensions → separate Date tables or USERELATIONSHIP
"Previous year" / "prior period" / "YTD"	Time Intelligence → SAMEPERIODLASTYEAR / TOTALYTD / DATEADD
Text cleanup / column splitting / profiling	Power Query → match function to the required output
"Does this meet the goal?" (Yes / No)	Evaluate ALL requirements — partial compliance = No
Fact + dimension + relationship diagram	Modelling → check cardinality, direction, and active/inactive
Dataset is slow	Performance → star schema, aggregations, Dual storage, reduce cardinality

Each user sees only their own data	Row-Level Security → Dynamic RLS using USERNAME()
Running total or cumulative figure	CALCULATE + TOTALYTD or DATESYTD
Multiple files in a folder	Folder connector (not the individual file-type connector)

What Is Coming in Session 2

Session 2: DAX & Exam Strategy · 2.5 hours · 6 parts

Session 2 — Time Plan

Part	Topic
Part 1 (75 min)	DAX Measures & Calculations
Part 2 (20 min)	Time Intelligence
Part 3 (20 min)	Visualise & Analyse Data
Part 4 (15 min)	Security — RLS
Part 5 (15 min)	Power BI Service
Part 6 (5–10 min)	Performance Optimisation
Wrap-Up (10 min)	DAX Cheat Sheet + Exam Tips

Why Session 2 Matters — The Marks

Topic	Approx Marks Value
DAX (CALCULATE, FILTER, ALL, RELATED, SUMX)	~90 marks
Time Intelligence (SAMEPERIODLASTYEAR, TOTALYTD, DATEADD)	~40 marks
Visuals, RLS, Service, Performance (combined)	~160 marks
Session 2 total coverage	~290 marks

💡 **The Full Picture Sessions 1 and 2 together cover the full PL-300 exam.**

Session 1 covered Power Query and Data Modelling (~250 marks). Session 2 covers DAX, Time Intelligence, and the remaining domains (~290 marks).

Part 1 — DAX Measures & Calculations

The functions you need to know:

Function / Concept	What It Does

CALCULATE	Evaluates an expression in a modified filter context — the most important DAX function in the exam
FILTER	Returns a filtered table — used inside CALCULATE to define row-level conditions
ALL	Removes all filters from a table or column — used to ignore slicers or create % of total
ALLSELECTED	Removes model filters but keeps filters the user applied in visuals or slicers
RELATED	Pulls a value from a related dimension table into the current row context
SUMX	Iterates row by row over a table and sums the result of an expression
DIVIDE	Safe division — returns blank or an alternate value instead of an error when dividing by zero
Context Transition	When a measure is called inside an iterator (e.g. SUMX), row context converts to filter context

Part 2 — Time Intelligence

Requirement	Function to Use
Sales from the same period last year	SAMEPERIODLASTYEAR
Shift a date range by N periods	DATEADD
Sales from the previous month	PREVIOUSMONTH
Sales from the previous year	PREVIOUSYEAR
Running year-to-date total	TOTALYTD
Running quarter-to-date total	TOTALQTD
Running month-to-date total	TOTALMTD

Part 3 — Visualise & Analyse Data

Feature	What Microsoft Tests
Slicers	Filter a single page — default behaviour
Sync Slicers	Filter across multiple pages simultaneously
Drillthrough	Right-click navigation to a detail page filtered by the selected value
Tooltips	Hover to see extra information — can use a custom report page as a tooltip
Conditional Formatting	Colour-code values in a table or matrix based on rules or a measure
Bookmarks	Save a view state — used for navigation buttons and toggle effects

Part 4 — Security: RLS

Concept	What to Know
Static RLS	Hardcoded filter per role (e.g. [Region] = "North"). Not scalable for large user bases.
Dynamic RLS	Filter uses USERPRINCIPALNAME() to match the logged-in user to a row in a security table. One role scales to all users.
USERPRINCIPALNAME()	Returns the Azure AD email of the current user — used in environments with Azure AD authentication
Azure AD Users	Users are assigned to roles in Power BI Service. Desktop only simulates roles for testing.

Part 5 — Power BI Service

Concept	What to Know
Workspaces	Collaboration containers for reports and datasets. My Workspace = personal. App Workspaces = team.
Apps	Published, read-only content bundles distributed to end users — separate from the editing workspace.
Dashboards vs Reports	Reports = multi-page, built in Desktop or Service. Dashboards = single-page tiles pinned from reports. Dashboards cannot be built in Desktop.
Scheduled Refresh	Automatic data refresh — requires a gateway for on-premises data sources.
Gateways	Bridges Power BI Service to on-premises sources. Required for scheduled refresh of SQL Server, local Excel files, etc.

Part 6 — Performance Optimisation

Concept	What to Know
Import vs DirectQuery	Import = cached, fast, needs refresh. DirectQuery = live, always current, slower. Choose Import by default.
Dual Storage Mode	Dimension tables in composite models — behaves as Import alone, DirectQuery alongside DirectQuery tables.
Star Schema	Flat tables = slow. Star schema = fast. Reduces cardinality and improves compression.
Aggregations	Pre-summarised tables Power BI uses instead of scanning the full fact table for high-level queries.

The 6 Exam Patterns — How Session 2 Is Organised

💡 **Key Insight Session 2 is organised around 6 exam patterns, not 6 feature lists.**

Microsoft tests requirements, not feature definitions. If you can spot the pattern in a question, you already know the answer category.

When the question says…	The pattern is…
"Need a calculation that changes based on context"	DAX — CALCULATE + FILTER
"Need previous period / running total / YTD"	Time Intelligence — match requirement to function
"Need users to interact with the report"	Visuals — slicer, drillthrough, tooltip, bookmark
"Need users to see only their own data"	Security — Dynamic RLS + USERPRINCIPALNAME()
"Need to share, deploy, or refresh"	Power BI Service — workspace, app, gateway
"Dataset is slow / report takes too long"	Performance — star schema, Import, Dual, aggregations

Your Preparation Task Before Session 2

📝 **Your Homework Task 1 — DAX:**

Write one sentence for each function: CALCULATE, FILTER, ALL, ALLSELECTED, RELATED, SUMX, DIVIDE. What does it do? When would you use it?

Task 2 — Time Intelligence:

Memorise the map: SAMEPERIODLASTYEAR (last year), PREVIOUSMONTH (last month), TOTALYTD (year running total), DATEADD (shift by N periods), TOTALQTD / TOTALMTD (quarter and month running totals).

Task 3 — Patterns:

Learn the 6 exam patterns by name. In Session 2, every question will be introduced by its pattern first.

Target: score 75%+ on any practice set before Session 2.

PL-300 Exam Preparation

Session 2: DAX & Exam Strategy

Learner Guide · Exam Strategy Workshop

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> **How to Use This Guide**
> This guide is your standalone reference for Session 2. Every concept, formula, and table in the session is recorded here so you can revise without needing the slides or the trainer present. Work through the practice questions from the bank alongside this guide.

What This Session Covers

Part	Topic
Part 1	DAX Measures & Calculations
Part 2	Time Intelligence
Part 3	Visualise & Analyse Data
Part 4	Security — Row-Level Security
Part 5	Power BI Service
Part 6	Performance Optimisation
Wrap-Up	DAX Cheat Sheet + Exam Tips

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Session 2 total coverage	~290 marks

> **The Full Picture**
> Session 1 covered Power Query and Data Modelling (~250 marks). Session 2 covers DAX, Time Intelligence, and the remaining domains (~290 marks). Together, both sessions cover the complete PL-300 exam.

The 6 Exam Patterns — Your Mental Framework

Before every question in this session, identify which of these six patterns applies. Once you know the pattern, you know the answer category.

When the question says…	The pattern is…
"Need a calculation that changes based on context"	DAX — CALCULATE + FILTER
"Need previous period / running total / YTD"	Time Intelligence — match to function
"Need users to interact with the report"	Visuals — slicer, drillthrough, tooltip, bookmark
"Need users to see only their own data"	Security — Dynamic RLS + USERPRINCIPALNAME()
"Need to share, deploy, or refresh"	Service — workspace, app, gateway
"Dataset is slow / report takes too long"	Performance — star schema, Import, Dual, reduce columns

Part 1

DAX Measures & Calculations

Questions for this part

Q25, Q53, Q57, Q65, Q66, Q148, Q161, Q163, Q165, Q166, Q181, Q182, Q186, Q187, Q201, Q213, Q220, Q223, Q246, Q284

How Microsoft Tests DAX

The PL-300 does not ask you to write DAX from scratch. It asks you to:

- Choose the correct function for a described business requirement
- Complete a partially written formula — HOTSPOT or "Complete the Solution" questions
- Identify why a formula produces the wrong result
- Understand filter context — why the same measure returns different numbers in different visuals

Key Insight

Every DAX question on the PL-300 is really a filter context question in disguise. Once you understand how context works, you understand DAX.

CALCULATE — The Most Important DAX Function

CALCULATE does two things simultaneously: it evaluates an expression AND it changes the filter context in which that expression is evaluated. It is the most-tested DAX function on the PL-300.

Syntax

```
CALCULATE ( <expression>, <filter1>, <filter2>, ... )
```

Arguments

Argument	What It Does
expression	The measure or aggregate to evaluate (e.g. [Total Revenue], SUM(Sales[Amount]))
filter1, filter2…	The new filter conditions. These REPLACE (not add to) the existing filter for those columns.

Why You Use It

Without CALCULATE you cannot override the filter context that a slicer, row, or column imposes on a measure. CALCULATE is how you answer questions like "show me last year's revenue regardless of what year is selected" or "show me total revenue for the South region only."

Common Patterns

```
-- Override the year filter (hardcoded year):
Revenue 2023 =
CALCULATE ( [Total Revenue], DateTable[Year] = 2023 )

-- Remove all filters on Sales (grand total):
Total Revenue All =
CALCULATE ( [Total Revenue], ALL ( Sales ) )

-- Apply a row-level filter:
South Revenue =
CALCULATE ( [Total Revenue], State_Regions[Region] = "South" )
```

> **Common Pitfall**
> Q57 trap: a measure with CALCULATE and a hardcoded year filter will IGNORE the year slicer on the report page. CALCULATE replaces the filter on that column — it does not add to it. This surprises learners who expect the slicer to still apply.

FILTER — Row-Level Filtering Inside CALCULATE

What It Does

FILTER returns a table containing only the rows that pass a condition. Used inside CALCULATE when the condition involves a computed value or a value from a related table — not a simple column equality.

Syntax

```
FILTER ( <table>, <condition> )
```

Worked Example

```
-- Revenue from orders with quantity greater than 10:
High Volume Revenue =
CALCULATE (
    [Total Revenue],
    FILTER ( Sales, Sales[Order Quantity] > 10 )
)
```

Important Distinction

FILTER iterates every row of the table — it is slower than a simple column filter argument. Use a column filter (e.g. Sales[Region] = "South") when possible. Use FILTER when the condition computes a value row by row (e.g. comparing quantity to a threshold).

ALL / ALLSELECTED / ALLEXCEPT — Removing Filters

The Three Variations

Function	What It Removes / Keeps
ALL (Table)	Removes ALL filters on the table. Grand total denominator. Ignores slicers.
ALL (Table[Column])	Removes filters on ONE column only. Other column filters still apply.
ALLSELECTED (Table)	Removes model filters but KEEPS filters the user applied via slicers or visuals. "% of selected total."
ALLEXCEPT (Table, Col1, Col2)	Removes all filters EXCEPT on the specified columns. Used in security and running total patterns.

The % of Total Pattern (Q53)

```
-- Revenue as % of grand total (ignores ALL slicers):
Revenue % of Total =
DIVIDE ( [Total Revenue], CALCULATE ( [Total Revenue], ALL ( Sales ) ) )


-- Revenue as % of slicer selection (respects user slicers):
Revenue % of Selected =
DIVIDE ( [Total Revenue], CALCULATE ( [Total Revenue], ALLSELECTED ( Sales
) ) )
```

Exam Decision Rule

"% of total that ignores slicers" → ALL. "% of total that updates when the user uses a slicer" → ALLSELECTED. This distinction appears in multiple PL-300 questions.

RELATED — Pulling from a Dimension Table

What It Does

RELATED retrieves a single value from a related table using the existing relationship. It works in calculated columns only — it cannot be used in measures.

Syntax

```
RELATED ( <RelatedTable[Column]> )
```

Worked Example

```
-- Add the Region from State_Regions to the Sales table:
-- (Calculated column in the Sales table)
Sale Region =
RELATED ( State_Regions[Region] )
```

Common Pitfall

Q163 trap: RELATED only works in a calculated COLUMN, not in a measure. In a measure, use CALCULATE with FILTER and RELATED() together, or use LOOKUPVALUE(). If the exam asks for a measure that pulls from a related table, RELATED alone is wrong.

SUMX — Row-by-Row Iteration

What It Does

SUMX iterates over every row of a table, evaluates an expression for each row, and sums the results. It is required whenever multiplication happens at the row level before summing — for example, calculating revenue as Quantity × Price per order line.

Syntax

```
SUMX ( <table>, <expression> )
```

Why SUM is Wrong for Revenue (Q25)

```
-- WRONG — multiplies totals, not row values:
Wrong Revenue = SUM ( Sales[Order Quantity] ) * SUM ( Sales[Product Sales
Price] )


-- CORRECT — multiplies at the row level then sums:
Total Revenue =
SUMX ( Sales, Sales[Order Quantity] * Sales[Product Sales Price] )
```

Key Concept

The difference matters when discount or other row-level adjustments exist. SUMX gives the correct result because it processes one order line at a time before summing. SUM gives the wrong result because it aggregates both columns first.

DIVIDE — Safe Division

What It Does

DIVIDE performs division and handles the case where the denominator is zero or blank — returning blank (or an alternate value) instead of an error. Always use DIVIDE instead of the / operator in measures.

Syntax

```
DIVIDE ( <numerator>, <denominator> [, <alternate_result>] )
```

Worked Example

```
-- Profit margin — safe if COGS is ever zero:
Profit Margin % =
DIVIDE ( [Total Revenue] - [Total COGS], [Total Revenue] )
```

Common Pitfall

Q165: using / instead of DIVIDE causes an error or Infinity when any denominator row is zero. DIVIDE returns BLANK, which visuals treat as "no value" — a much cleaner result.

VAR / RETURN — Variables in DAX

What It Does

VAR stores an intermediate result so it can be reused in the same formula. RETURN specifies the final value the measure produces. Variables make complex formulas readable and prevent the same calculation from running twice.

Syntax

```
Measure Name =
VAR VariableName = <expression>
RETURN
    <expression using VariableName>
```

Worked Example (Q186, Q223)

```
Revenue YoY Growth % =
VAR CurrentRevenue = [Total Revenue]
VAR PriorRevenue = CALCULATE ( [Total Revenue], SAMEPERIODLASTYEAR (
DateTable[Date] ) )
RETURN
    DIVIDE ( CurrentRevenue - PriorRevenue, PriorRevenue )
```

Key Concept

Without VAR, the prior year measure would need to be written twice inside the DIVIDE — once in the numerator and once in the denominator — and Power BI would evaluate it twice. VAR computes it once, stores the result, and reuses it.

RANKX — Ranking a Measure

What It Does

RANKX ranks a measure across every row of a table. The most common use: rank every product by revenue, every region by profit, or every customer by order count.

Syntax

```
RANKX ( <table>, <expression>, [<value>], [<order>], [<ties>] )
```

Argument	What It Controls
table	The table to rank across. Always wrap in ALL() to remove the current filter — otherwise every row ranks 1 against itself.
expression	The measure to rank by (e.g. [Total Revenue])
order	DESC = highest value = rank 1. ASC = lowest value = rank 1.
ties	Dense = no gaps (1,2,3 even if two items tie). Skip = standard competition ranking (1,1,3 if two items tie at rank 1).

Worked Example (Q187, Q220)

```
Product Revenue Rank =
RANKX ( ALL ( Products[Product Name] ), [Total Revenue], , DESC, Dense )
```

Common Pitfall

Q220: omitting ALL() causes every product to rank 1 against itself in the current filter context. ALL() removes the filter so the ranking is calculated across ALL products simultaneously.

COUNTROWS — Counting Rows

What It Does

COUNTROWS counts the number of rows in a table. Combined with CALCULATE and FILTER, it counts the number of rows meeting a condition.

Worked Example (Q66)

```
-- Count orders where a discount was applied:
Discounted Orders =
CALCULATE (
    COUNTROWS ( Sales ),
    NOT ISBLANK ( Sales[Discount Applied] )
)
```

Context Transition — Why Iterators Work

The Concept

When a measure is called inside an iterator function (SUMX, AVERAGEX, MAXX, etc.), the row context of the iterator automatically converts to a filter context for the measure. This is called context transition.

This is why SUMX can call a measure in its expression and still get the correct result for each row — the iterator creates a row context, which becomes a filter context the moment it touches the measure.

Exam Pattern (Q284)
Q284 is a Yes/No question: "A measure is called inside SUMX. Does the measure see each row individually?" Answer: Yes — because context transition converts the row context to a filter context for each row.

USERELATIONSHIP Inside CALCULATE

What It Does

USERELATIONSHIP activates an inactive relationship for a specific CALCULATE expression. It is used when a fact table has two date columns (e.g. Order Date and Delivery Date) both linked to the DateTable — only one relationship can be active at a time.

Worked Example (Q65)

```
-- Revenue calculated using the Delivery Date instead of the Order Date:
Delivery Revenue =
CALCULATE (
    [Total Revenue],
    USERELATIONSHIP ( Sales[DeliveryDate], DateTable[Date] )
)
```

Common Pitfall
Q65 trap: learners delete the active relationship to make room for the new one. The correct approach is always to create an INACTIVE relationship and activate it with USERELATIONSHIP inside CALCULATE. Deleting the active relationship breaks all existing measures.

Part 2
Time Intelligence

Questions for this part
Q34, Q38, Q45, Q56, Q63, Q183, Q184, Q185, Q211, Q236

Prerequisite — The Date Table

All time intelligence functions require a Date table that meets these three conditions:

- Contiguous — no missing dates in the range (1 Jan to 31 Dec for every year in the data)
- Marked as a Date Table in Power BI (Table tools → Mark as Date Table)
- Related to the fact table via a one-to-many relationship on the Date column

Important
Without "Mark as Date Table," functions like TOTALYTD and SAMEPERIODLASTYEAR will produce incorrect results or errors. This step is tested directly in the question bank (Q35, Q64, Q142).

The Time Intelligence Function Map — Memorise This

Business Requirement	Function to Use
Sales from the same period last year	SAMEPERIODLASTYEAR (DateTable[Date])
Shift a date range by N periods (flexible)	DATEADD (DateTable[Date], -1, YEAR/QUARTER/MONTH)
Sales from the previous month	PREVIOUSMONTH (DateTable[Date])
Sales from the previous year (full calendar year)	PREVIOUSYEAR (DateTable[Date])
Running year-to-date total	TOTALYTD ([Measure], DateTable[Date])
Running quarter-to-date total	TOTALQTD ([Measure], DateTable[Date])
Running month-to-date total	TOTALMTD ([Measure], DateTable[Date])
Parallel period comparison (any interval)	PARALLELPERIOD (DateTable[Date], -1, YEAR)

SAMEPERIODLASTYEAR — Most Common Pattern

What It Does

Returns the same period from the previous year. If the current filter is June 2024, SAMEPERIODLASTYEAR returns June 2023. Used inside CALCULATE.

Syntax

```
CALCULATE ( <expression>, SAMEPERIODLASTYEAR ( <DateColumn> ) )
```

Worked Examples (Q34, Q38, Q56)

```
-- Revenue for the same period last year:
Revenue LY =
CALCULATE ( [Total Revenue], SAMEPERIODLASTYEAR ( DateTable[Date] ) )


-- Year-on-year growth % (Q63):
Revenue YoY % =
VAR CurrentYear = [Total Revenue]
VAR PriorYear = CALCULATE ( [Total Revenue], SAMEPERIODLASTYEAR (
DateTable[Date] ) )
RETURN
    DIVIDE ( CurrentYear - PriorYear, PriorYear )
```

TOTALYTD — Running Year-to-Date Total

What It Does

TOTALYTD accumulates a measure from 1 January of the current year to the current date in the filter context. Each month's YTD value includes all months before it in the same year.

Syntax

```
TOTALYTD ( <expression>, <DateColumn> [, <year_end_date>] )
```

Worked Example (Q183, Q236)

```
Revenue YTD =
TOTALYTD ( [Total Revenue], DateTable[Date] )
```

> **Common Pitfall**
> Do not confuse TOTALYTD with SAMEPERIODLASTYEAR. TOTALYTD accumulates from 1 Jan to now — a running total. SAMEPERIODLASTYEAR shifts the date window back one year — a comparison point. These are different answers to different questions.

DATEADD — Flexible Period Shifting

What It Does

DATEADD shifts a set of dates by a specified number of periods (years, quarters, months, or days). Used when SAMEPERIODLASTYEAR does not cover the required shift — for example, "previous quarter" or "shift back 3 months."

Syntax

```
CALCULATE ( <expression>, DATEADD ( <DateColumn>, <number_of_periods>,
<interval> ) )
```

```
-- interval options: YEAR, QUARTER, MONTH, DAY
-- negative number = shift back; positive = shift forward
```

Worked Example (Q45, Q211)

```
-- Revenue for the previous quarter:
Revenue Previous Quarter =
CALCULATE ( [Total Revenue], DATEADD ( DateTable[Date], -1, QUARTER ) )

-- Revenue from 6 months ago:
Revenue 6M Ago =
CALCULATE ( [Total Revenue], DATEADD ( DateTable[Date], -6, MONTH ) )
```

Exam Decision Rule
"Previous quarter" → DATEADD with -1, QUARTER. "Same period last year" → SAMEPERIODLASTYEAR (simpler, no arguments needed). When in doubt: SAMEPERIODLASTYEAR for year comparisons; DATEADD for everything else.

PREVIOUSMONTH / PREVIOUSYEAR

What They Do

These functions return the complete previous month or year regardless of the current day. Unlike SAMEPERIODLASTYEAR (which returns the matching period of the prior year), PREVIOUSYEAR returns the entire prior calendar year.

```
-- All of last month:
Revenue Previous Month =
CALCULATE ( [Total Revenue], PREVIOUSMONTH ( DateTable[Date] ) )

-- All of last year:
Revenue Previous Year =
CALCULATE ( [Total Revenue], PREVIOUSYEAR ( DateTable[Date] ) )
```

Exam Tip (Q184, Q185)
Q184 and Q185 are HOTSPOT questions — you select the correct function from a dropdown. The trick: "previous month" → PREVIOUSMONTH; "running year total" → TOTALYTD; "shift by 2 quarters" → DATEADD. Match the phrase to the function exactly.

Part 3

Visualise & Analyse Data

Questions for this part
Q37, Q40, Q42, Q43, Q44, Q46, Q49, Q50, Q54, Q78

How Microsoft Tests Visuals

Visuals questions on the PL-300 do not test which visual looks best. They test which feature serves a specific user interaction requirement. The answer almost always comes down to one keyword in the scenario:

Keyword in the scenario	Feature to use
"User wants to filter across all pages"	Sync Slicers
"User right-clicks and navigates to a detail page"	Drillthrough
"User hovers to see extra information"	Tooltip (or custom page tooltip)
"User clicks a button to toggle between views"	Bookmarks + Buttons
"User expands a hierarchy within the same chart"	Drill-down (not drillthrough)
"Show which factors most influence a metric"	Key Influencers AI Visual
"User asks a question in natural language"	Q&A Visual
"Show colour-coded values based on a rule"	Conditional Formatting

Feature Reference — Full Coverage

Slicers and Sync Slicers

A slicer is a visual filter control that the user interacts with. By default, a slicer filters only the page it is on.

Sync Slicers (View → Sync slicers) allow the same slicer to filter multiple pages simultaneously. Each page can be set to sync the slicer value, show the slicer visually, or both — independently.

Common Pitfall
Q175 pattern: the requirement is "a slicer on Page 1 should also filter Page 2 and Page 3." The answer is Sync Slicers, not adding the same slicer to every page manually. Adding it manually creates three independent slicers, not one synchronised slicer.

Drillthrough vs Drill-down vs Tooltip

Feature	What Happens
Drill-down	Expands a hierarchy one level within the SAME visual. Year → Quarter → Month on a line chart.

	User clicks the drill-down button (arrow icons at the top of the visual).
Drillthrough	Navigates to a DIFFERENT page. User right-clicks a data point → Drillthrough → [page name]. The target page must have a drillthrough field configured in its Visualizations pane.
Tooltip (field)	A small pop-up appears on hover showing an extra field value. Added via the Tooltip well in the visual's field configuration.
Tooltip (page)	A full report page appears on hover. Set the target page's Page information → Tooltip = On. Then set the source visual's Tooltip to that page.

Bookmarks and Buttons (Q78)

A bookmark captures the current state of a report page — which visuals are visible, which filters are applied, and what values slicers hold. Bookmarks are assigned to buttons to enable toggle effects and navigation.

Q78 scenario: a button on the report should show one chart when clicked and a different chart when clicked again. The correct solution:

- Step 1: Create two visuals on the same page. Show only the first, hide the second. Create Bookmark A.
- Step 2: Show only the second, hide the first. Create Bookmark B.
- Step 3: Add a button. Set Action → Bookmark → Bookmark A on the first button, Bookmark B on the second.

Common Pitfall

Learners choose drillthrough instead of bookmarks when "button" appears in the question. Drillthrough navigates to a different page. Bookmarks change the view on the current page.

Conditional Formatting

Conditional formatting applies colour, icons, data bars, or web URLs to cells in a table or matrix based on rules or a measure. Applied via: select the visual → Format → Conditional formatting → select the field.

Type	When to Use
Background colour (rules)	Colour-code cells by value range (e.g. red if revenue < target, green if above)
Background colour (measure)	Use a DAX measure to control the colour value dynamically
Data bars	Show a bar inside each cell proportional to its value
Icons	Show a traffic-light icon set based on rules
Web URL	Make a cell a clickable hyperlink — requires a column containing a URL

Analytics Pane (Q42, Q49)

The Analytics pane (the magnifier icon in the Visualizations pane) adds reference lines to supported visuals. Available line types:

- Constant line — a fixed value line (e.g. a target of 500,000)
- Min / Max / Average / Median line — computed automatically from the data
- Percentile line — a line at a specified percentile
- Forecast — project future values on a line chart (time-series data required)

> **Exam Trap**
> Q42: "Add a median reference line to the chart." The answer is Analytics pane → Median line. Not a new measure, not conditional formatting. The Analytics pane is the only way to add a reference line without writing DAX.

AI Visuals (Q44, Q50)

AI Visual	When to Use It
Key Influencers	"Which factors most increase or decrease [metric]?" — the visual analyses the data and lists the top drivers with their relative impact.
Decomposition Tree	"Let me break down [metric] by any dimension interactively." — the user selects the next dimension to drill into at runtime.
Q&A Visual	"Users want to ask questions in natural language." — the visual accepts typed questions and generates a chart or number in response.
Smart Narrative	"Automatically generate a text summary of the key data points in the report." — the visual writes a paragraph describing the data.

Part 4

Security — Row-Level Security

Questions for this part
Q51, Q74, Q167, Q196, Q226

What is Row-Level Security (RLS)?

Row-Level Security restricts which rows of data a user can see when they view a report. It is applied through roles — each role has a DAX filter that determines which rows are visible to users assigned to that role.

Static RLS vs Dynamic RLS

Type	How It Works
Static RLS	A DAX filter is hardcoded per role. Example: [Region] = "North". One role = one region. To cover 50 regions you need 50 roles. Scalable only for a small number of fixed groups.
Dynamic RLS	The DAX filter uses USERPRINCIPALNAME() or USERNAME() to match the logged-in user to a row in a security table. One role covers all users — scales automatically.

Dynamic RLS — The Standard Exam Pattern

Step 1: Create a Security Table

The security table has one row per user and maps each user to their allowed dimension value (e.g. region, department, store).

Email	Region
alice@company.com	South
bob@company.com	North
carol@company.com	West

Step 2: Relate the Security Table to the Model

Create a relationship between the Security table's Region column and the region column in your data model (e.g. State_Regions[Region]).

Step 3: Create One Role with a DAX Filter

In Power BI Desktop: Model tab → Manage roles → Create role → Apply a DAX filter to the Security table:

```
-- DAX filter on the Security table (Email column):
[Email] = USERPRINCIPALNAME()
```

Step 4: Assign Users in Power BI Service

In Power BI Service: go to the dataset → Security → assign users to the role. All users go into the same single role — the DAX filter handles which rows each user sees.

Key RLS Facts for the Exam

Topic	The Rule
Where roles are defined	In Power BI Desktop — Model tab → Manage roles
Where users are assigned	In Power BI Service — Dataset → Security → add users to roles. NEVER in Desktop.
Testing RLS	In Desktop: Model tab → View as (simulates a role). In Service: Test as role.
USERPRINCIPALNAME()	Returns the Azure AD email address of the current user. Use this in cloud / Azure AD environments.
USERNAME()	Returns domain\username. Use in on-premises / Windows authentication environments.
Static vs Dynamic	Static = hardcoded filter per role. Dynamic = USERPRINCIPALNAME() in the filter. Dynamic scales; static does not.
OLS — Object-Level Security	Hides entire tables or columns from a role. Configured in Tabular Editor, not the Desktop RLS UI. Know the name but not the full setup detail.

Common Pitfall — Q74

Q74 (Yes/No): the proposed solution assigns users to roles in Power BI Desktop. Answer: No. Users must be assigned in Power BI Service. Desktop only supports testing roles, not managing user assignments.

Common Pitfall — Q51

Q51: the scenario describes 50 regions and says new regions are added frequently. Static RLS with one role per region requires a developer to update roles every time a new region is added. Dynamic RLS handles new regions automatically — no role update needed.

Part 5
Power BI Service

Questions for this part
Q8, Q9, Q47, Q68, Q70, Q72, Q77, Q95

Power BI Service — Full Reference

Workspaces

Workspaces are containers for Power BI content (reports, datasets, dashboards). There are two types:

- My Workspace — personal only. Cannot be shared with others. No team collaboration.
- App Workspace (shared workspace) — for teams. Members collaborate on the same content. Content can be published as an App.

Workspace Roles

Role	Key Permissions
Admin	Full control — add/remove members, delete workspace, publish apps, update all content
Member	Publish content, share items, allow others to reshare content. Cannot add workspace members.
Contributor	Create, edit, and delete workspace content. Cannot add members or share content.
Viewer	View and interact with published content. Read-only access.

Least Privilege Rule
When the exam asks for the minimum permission level to achieve a requirement, always choose the role that grants exactly what is needed and no more. Q72: need to edit reports but not add members → Contributor (not Member, which over-grants).

Apps

An App is a published, read-only bundle of workspace content (reports, dashboards, datasets) distributed to end users. Key facts:

- Apps are separate from the editing workspace — end users see the app, not the workspace itself
- Workspace members can continue editing the workspace content; app viewers see a stable published version
- The app must be explicitly updated and republished when workspace content changes

- App audiences allow different groups of users to see different content within the same app

Dashboards vs Reports

Reports	Dashboards
Multi-page	Single page (canvas)
Created in Desktop or Service	Created in Service only — cannot build dashboards in Desktop
Interactive — slicers, filters, drill	Limited interactivity — tiles link back to reports
Full DAX and modelling support	Tiles are pinned from reports, live pages, or entered manually
Supports drillthrough, bookmarks, tooltips	Supports data alerts on numeric tiles

Scheduled Refresh and Gateways

Scheduled refresh automatically refreshes Import mode datasets on a schedule. Key rules:

- DirectQuery datasets are always live — scheduled refresh does not apply
- Cloud sources (Azure SQL, SharePoint Online, Excel on OneDrive) do not require a gateway
- On-premises sources (SQL Server, local Excel/CSV, SSAS) always require an on-premises gateway

Data Source	Gateway Required?
SQL Server (on-premises)	Yes — on-premises data gateway
Azure SQL Database	No — cloud source
Excel on local drive	Yes — on-premises gateway
Excel on OneDrive / SharePoint Online	No — cloud source
SSAS (Analysis Services)	Yes — on-premises gateway (even for Live Connection)
Power BI Dataflow	No — dataflows refresh in Service

Deployment Pipelines

Deployment pipelines promote content through Development → Test → Production environments. Used when teams need to test report changes before releasing them to end users. A Power BI Premium feature. Q9 and Q198 test the correct use of deployment pipelines vs direct publication to a production workspace.

Dataflows

Dataflows are reusable Power Query transformations stored in Power BI Service. Multiple datasets can connect to the same dataflow — avoiding duplicated ETL logic across reports. Q123, Q124 test dataflows as the correct answer when "reusable shared data preparation" appears in the scenario.

Common Pitfall — Q95

Q95: a company uses SSAS (Analysis Services) with a Live Connection. Can they schedule a refresh without a gateway? Answer: No. Even a live connection to an on-premises SSAS server requires an on-premises gateway to bridge Service to the SSAS instance.

Part 6

Performance Optimisation

Questions for this part
Q35, Q85, Q108, Q154, Q330, Q447

Performance — Quick Reference

Concept	The Rule
Star Schema vs Flat Table	Flat tables are slow because repeated string values cannot be compressed efficiently. Star schema separates descriptive data into dimension tables, reduces cardinality, and enables VertiPaq compression. Always the first recommendation for a slow model.
Remove Unused Columns	Every column in the model consumes memory. Remove columns not used in reports in Power Query (before loading). High-cardinality columns — GUIDs, free text, full timestamps — are the worst offenders.
Import vs DirectQuery	Import mode is almost always faster — data is cached in memory. DirectQuery queries the source live and is limited by the source system's speed. Default to Import; use DirectQuery only when real-time data is a hard requirement.
Disable Auto Date/Time	Auto date/time creates a hidden date table for every date column in the model. With multiple date columns, this creates multiple hidden tables consuming memory. Disable it (Options → Data Load → uncheck Auto date/time) and use a single shared DateTable.
Aggregation Tables	Pre-summarised tables that Power BI queries for high-level requests instead of scanning the full fact table. Used for very large datasets (tens of millions of rows). The engine routes queries automatically.
Dual Storage Mode	Set dimension tables to Dual in composite models. Allows them to serve both Import and DirectQuery fact tables efficiently without an extra query to the source.
Reduce Cardinality	Replace high-cardinality columns with lower-cardinality equivalents. Replace a full datetime with separate Date and Time columns. Replace GUIDs with integer surrogate keys.

Key Concept — Q35
Q35 tests Auto date/time. The correct answer: disable Auto date/time and create a single DateTable that all date columns share. This eliminates the hidden tables and reduces model size significantly.

Key Concept — Q108

Q108 tests high-cardinality columns. A GUID column (unique identifier per row) has the highest possible cardinality — it cannot be compressed at all. The answer: remove it in Power Query. If the GUID is needed as a relationship key, replace it with an integer surrogate key.

Wrap-Up — Cheat Sheets to Keep

Cheat Sheet A — DAX Function Reference

Function	One-Line Rule
CALCULATE	Evaluates a measure inside a new filter context — the most important DAX function.
FILTER	Returns a filtered TABLE — used inside CALCULATE for row-level conditions.
ALL	Removes ALL filters on a table or column — grand total denominator, ignores slicers.
ALLSELECTED	Removes model filters but KEEPS user-applied slicer filters.
RELATED	Pulls a value from a related dimension into the current row — calculated columns only.
SUMX	Row-by-row iteration — multiply before summing. Use for line-level revenue calculations.
DIVIDE	Safe division — returns BLANK on zero denominator, not an error.
VAR / RETURN	Store an intermediate calculation in a variable; output with RETURN.
RANKX	Rank a measure across a table — always wrap the table argument in ALL().
COUNTROWS	Count rows in a table — use CALCULATE + FILTER to count rows meeting a condition.
USERELATIONSHIP	Activate an inactive relationship inside a CALCULATE expression.
Context Transition	Row context inside an iterator (SUMX) becomes filter context for any measure called within it.

Cheat Sheet B — Time Intelligence Map

Requirement	Function
"Same period last year"	SAMEPERIODLASTYEAR (DateTable[Date])
"Previous year" (full calendar year)	PREVIOUSYEAR (DateTable[Date])
"Previous month" (full calendar month)	PREVIOUSMONTH (DateTable[Date])
"Shift by N periods"	DATEADD (DateTable[Date], -N, YEAR/QUARTER/MONTH)
"Year-to-date running total"	TOTALYTD ([Measure], DateTable[Date])

"Quarter-to-date"	TOTALQTD ([Measure], DateTable[Date])
"Month-to-date"	TOTALMTD ([Measure], DateTable[Date])

Cheat Sheet C — 6 Exam Patterns Decoder

If the question says…	The answer category is…
"Calculation that changes based on context"	DAX — CALCULATE + appropriate filter function
"Previous year / YTD / prior period"	Time Intelligence — match requirement phrase to function
"Users interact with the report"	Visuals — slicer, drillthrough, tooltip, bookmark
"Users see only their own data"	Security — Dynamic RLS + USERPRINCIPALNAME()
"Share, deploy, or refresh the report"	Service — workspace, app, gateway, scheduled refresh
"Report is slow / dataset is large"	Performance — star schema, Import, Dual, remove columns

Cheat Sheet D — RLS Quick Reference

Topic	The Rule
Where roles are defined	Power BI Desktop — Model → Manage roles
Where users are assigned	Power BI Service — Dataset → Security. Never in Desktop.
Static RLS	Hardcoded filter per role. Does not scale to large or changing user bases.
Dynamic RLS	USERPRINCIPALNAME() filter. One role. Scales automatically.
USERPRINCIPALNAME()	Returns the Azure AD email of the logged-in user.

Cheat Sheet E — Service Quick Reference

Topic	The Rule
My Workspace	Personal only. Cannot share with others.
App Workspace	Team collaboration. Content can be published as an App.
Apps	Read-only published content bundle for end users. Separate from the editing workspace.
Dashboards	Service only — cannot build in Desktop. Single page of pinned tiles.

Gateway required	On-premises sources: SQL Server, local Excel, SSAS. Not required for cloud sources.
Scheduled refresh	Import mode only. DirectQuery is always live — refresh does not apply.
Deployment Pipelines	Dev → Test → Production promotion. Premium feature.

Three Exam-Day Tips

Remember on Exam Day

Tip 1 — Decode before you answer: identify the Requirement, the Constraint, and the Keyword before reading the answer options. The keyword almost always tells you the correct category.

Tip 2 — Eliminate, do not guess: on every question, eliminate at least two wrong answers before choosing. Wrong answers are usually "right feature, wrong scenario."

Tip 3 — Protect your time: allocate no more than 2.5 minutes per question. Flag difficult questions and return to them. For Case Study questions, read the full scenario before answering any linked question.

Your Preparation Checklist

Before You Book the Exam

✓ You can explain what CALCULATE does and why it is the most important DAX function

✓ You can match any time intelligence requirement to the correct function without looking at the map

✓ You can describe the difference between drillthrough and drill-down without hesitation

✓ You can explain Dynamic RLS and write the USERPRINCIPALNAME() filter pattern from memory

✓ You know which sources require a gateway and which do not

✓ You are consistently scoring 75%+ on practice question sets from the bank

✓ You can name all 6 exam patterns and say which feature each maps to